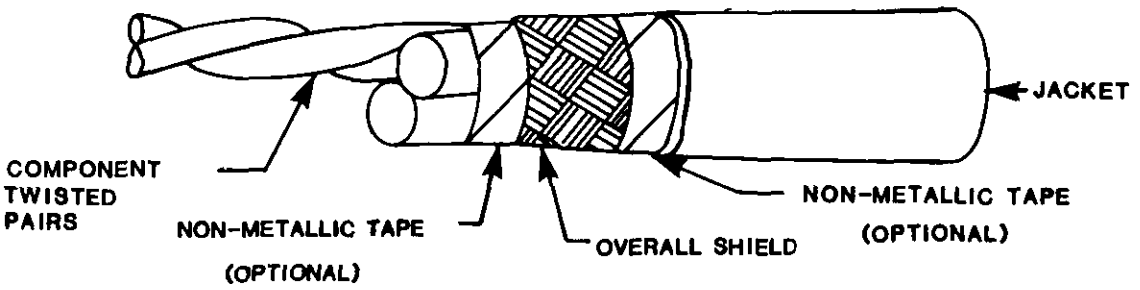
MIL-C-0024640/17A(SH)
23 October 1985
USED IN LIEU OF
MIL-C-24640/17
10 August 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, LIGHTWEIGHT, MULTI-PAIR, AWG 26, TYPE 2XOW

This military specification has been prepared by the Naval Sea Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-24640. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24640.



SH 12764

COMPONENT PAIRS:

First - Copper conductor, size AWG 26, 7 stranding. Nominal diameter of stranded conductor 0.020 inch.
Second - Material requirements in accordance with MIL-W-81044/12 except the conductor and identification codes and methods shall be in accordance with MIL-C-24640. Finished wire diameter 0.031 inch minimum and 0.037 inch maximum.

FIGURE 1. Construction (watertight).

(A) denotes changes.

FSC 6145

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(A) TABLE I. Construction details.

Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum)	Cable diameter (inch) (min)	Cable diameter (inch) (max)	Cable weight (pounds per 1000 feet) (max)
2XOW-6	6	0.229	0.038	0.336	0.363	107.0
2XOW-18	18	.374	.041	.468	.504	201.0
2XOW-24	24	.447	.044	.546	.588	266.0
2XOW-42	42	.567	.044	.646	.686	365.0
2XOW-60	60	.661	.047	.744	.802	485.0
2XOW-77	77	.770	.047	.840	.906	615.0

TABLE II. Pair lay.1/

Pair no.	Pair lay (inches)	Pair no.	Pair lay (inches)	Pair no.	Pair lay (inches)
1	1.5	27	1.0	53	2.5
2	1.0	28	2.0	54	3.0
3	2.0	29	3.0	55	4.5
4	3.0	30	4.5	56	5.5
5	4.5	31	1.0	57	1.0
6	1.0	32	2.0	58	1.5
7	2.0	33	3.0	59	2.0
8	3.0	34	4.5	60	2.5
9	1.5	35	1.0	61	3.0
10	2.5	36	2.0	62	4.5
11	5.5	37	3.0	63	5.5
12	1.5	38	4.5	64	1.0
13	2.5	39	1.0	65	1.5
14	3.7	40	2.0	66	2.0
15	5.5	41	3.0	67	2.5
16	1.5	42	4.5	68	3.0
17	2.5	43	1.0	69	4.5
18	3.7	44	1.5	70	5.5
19	1.5	45	2.0	71	1.0
20	2.5	46	2.5	72	1.5
21	3.7	47	3.0	73	2.0
22	5.5	48	4.5	74	2.5
23	1.0	49	5.5	75	3.0
24	2.0	50	1.0	76	4.5
25	3.0	51	1.5	77	5.5
26	4.5	52	2.0		

1/ All lays are left hand.

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TABLE III. Assembly.

Type number	Core	Layer 1	Assembly layer 2	Layer 3	Layer 4
2XOW-6	----	Pairs 1-6			
2XOW-18	Pair 1	Pairs 2-6	Pairs 7-18		
2XOW-24	Pairs 1-2	Pairs 3-10	Pairs 11-24		
2XOW-42	----	Pairs 1-8	Pairs 9-22	Pairs 23-42	
2XOW-60	Pair 1	Pairs 2-7	Pairs 8-19	Pairs 20-36	Pairs 37-60
2XOW-77	Pairs 1-3	Pairs 4-12	Pairs 13-27	Pairs 28-49	Pairs 50-77

REQUIREMENTS:

Qualification required.
Construction details: See table I.

Overall shield shall be tin-coated copper braid with a coverage not less than 85 percent determined in accordance with MIL-C-27500.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum	42.1
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2500
Conductor to shield	1000
Shield to shield	Not applicable
Insulation resistance - megohms/1000 feet, minimum	5000
Conductor and shield continuity	No failure
Jacket flaws	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	Not applicable
Capacitance unbalance - percent, maximum	Not applicable
Characteristic impedance - at 1 MHz, ohms	Not applicable
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	No failure
Weight	No failure

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group B:	
Cold bending	No failure
Drip	Zero
Tear - pounds per inch thickness, minimum	35
Physicals (unaged)	
Jacket	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
Gas flame (1 hour)	Not applicable
Group C:	
Physicals (aged)	
Jacket	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Permanence of printing (conductor) - cycles, minimum	50
Permanence of printing (jacket) - cycles, minimum	250
Armor - conformance to material, construction and coverage	Not applicable
Shield - conformance to material, construction and coverage	No failure
Heat distortion - percent unaged, maximum	30
Cable filler removability	No failure
Shrinkage	No failure
Mutual inductance - dB/1000 feet, maximum	Not applicable
Surface transfer impedance	
Milliohms per meter, maximum	700
EMP response, dB, minimum	60
Pulse response time - ms/500 feet	
Rise time, maximum	Not applicable
Decay time, maximum	Not applicable

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Requirements
Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation	No failure
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer - (type A) hardness, minimum	80
Weathering	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	Not applicable
Finished wire	Not applicable

Preparing activity:
Navy - SH
(Project 6145-N322-17)

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